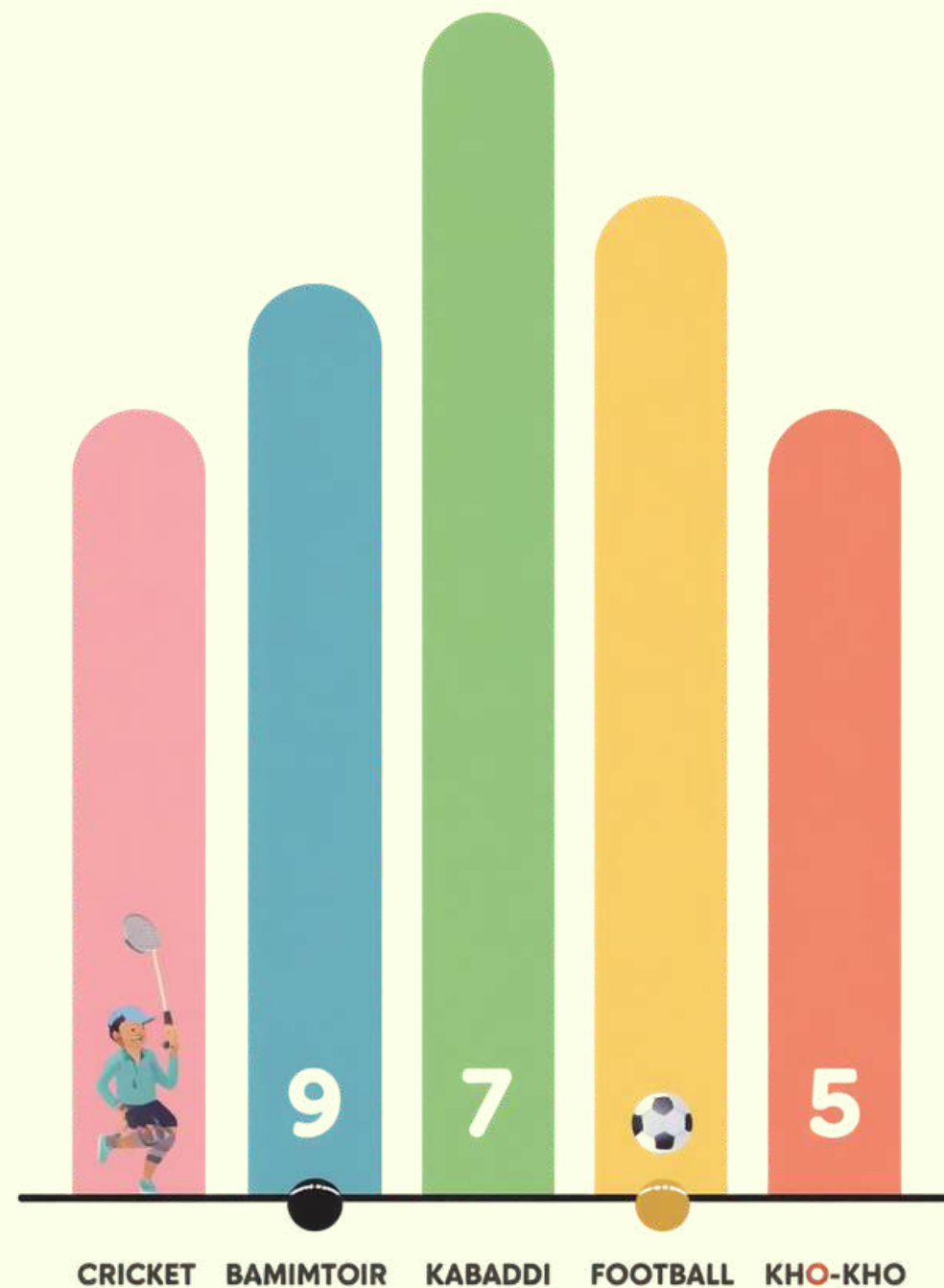




Python – Matplotlib: Plot Your Data

FAVOURITE SPORTS BAR CHART



Favourite Sports in Class 4

Someone typed 7 lines of code and
THIS appeared!
Which sport won? How can you tell?

Which Sport Won?

Cricket — 14 votes!

How do we know Cricket is the favourite sport? Look at the tallest bar on the chart!



What is Data?

Data is a collection of facts, numbers, or names. Shreya asked her friends how many glasses of water they drink daily:

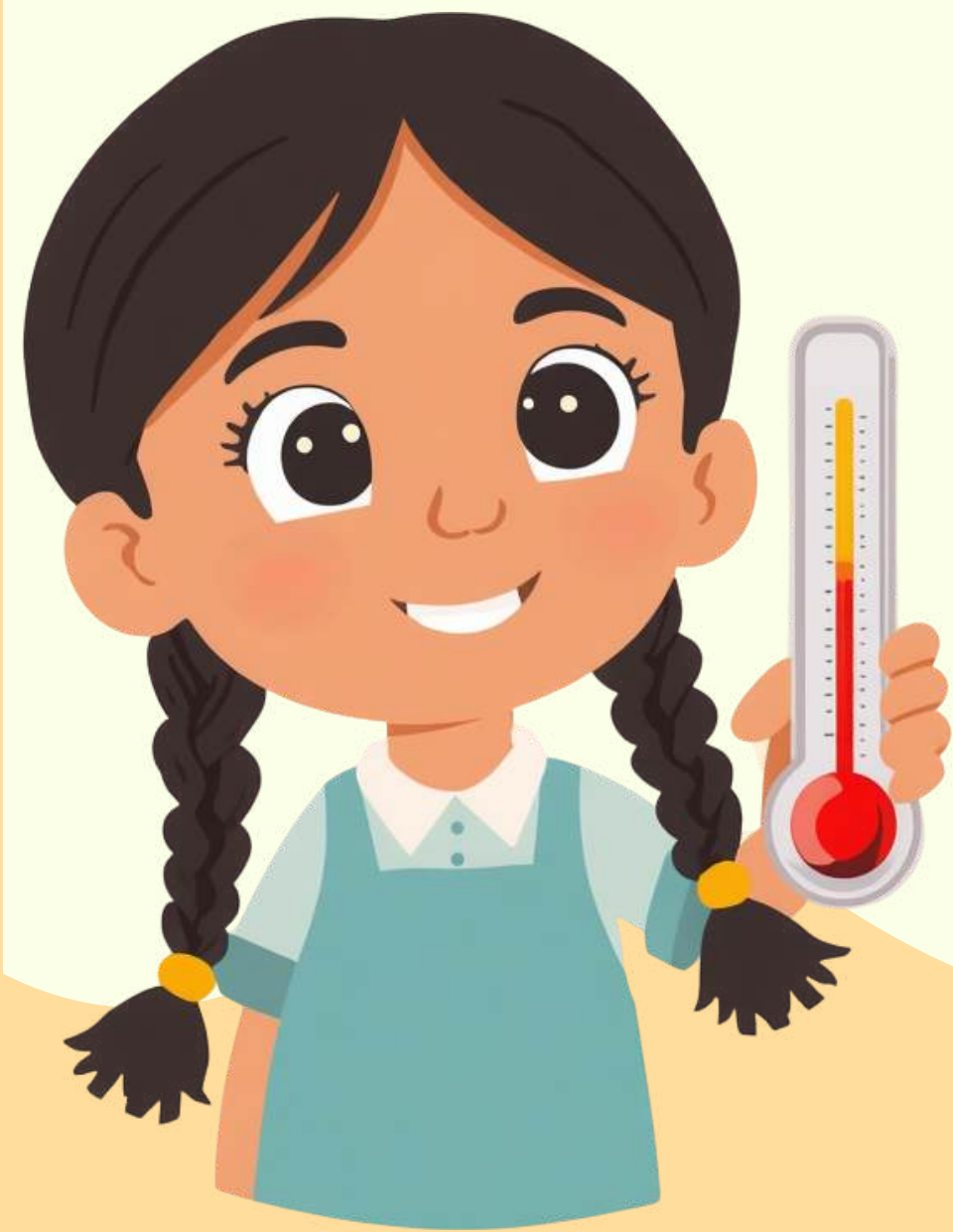
- Riya – 6 glasses
- Arjun – 4 glasses
- Meena – 8 glasses
- Karthik – 5 glasses
- Divya – 7 glasses



Where Do We See Data?

Data is all around us! Here are some examples:

- Test scores in your exams
- Temperature readings each day
- Cricket runs scored by players
- School attendance records
- Rainfall amounts in different cities



Why Visualize Data?

Our brain processes images 60,000x faster than text!

Compare rainfall data:

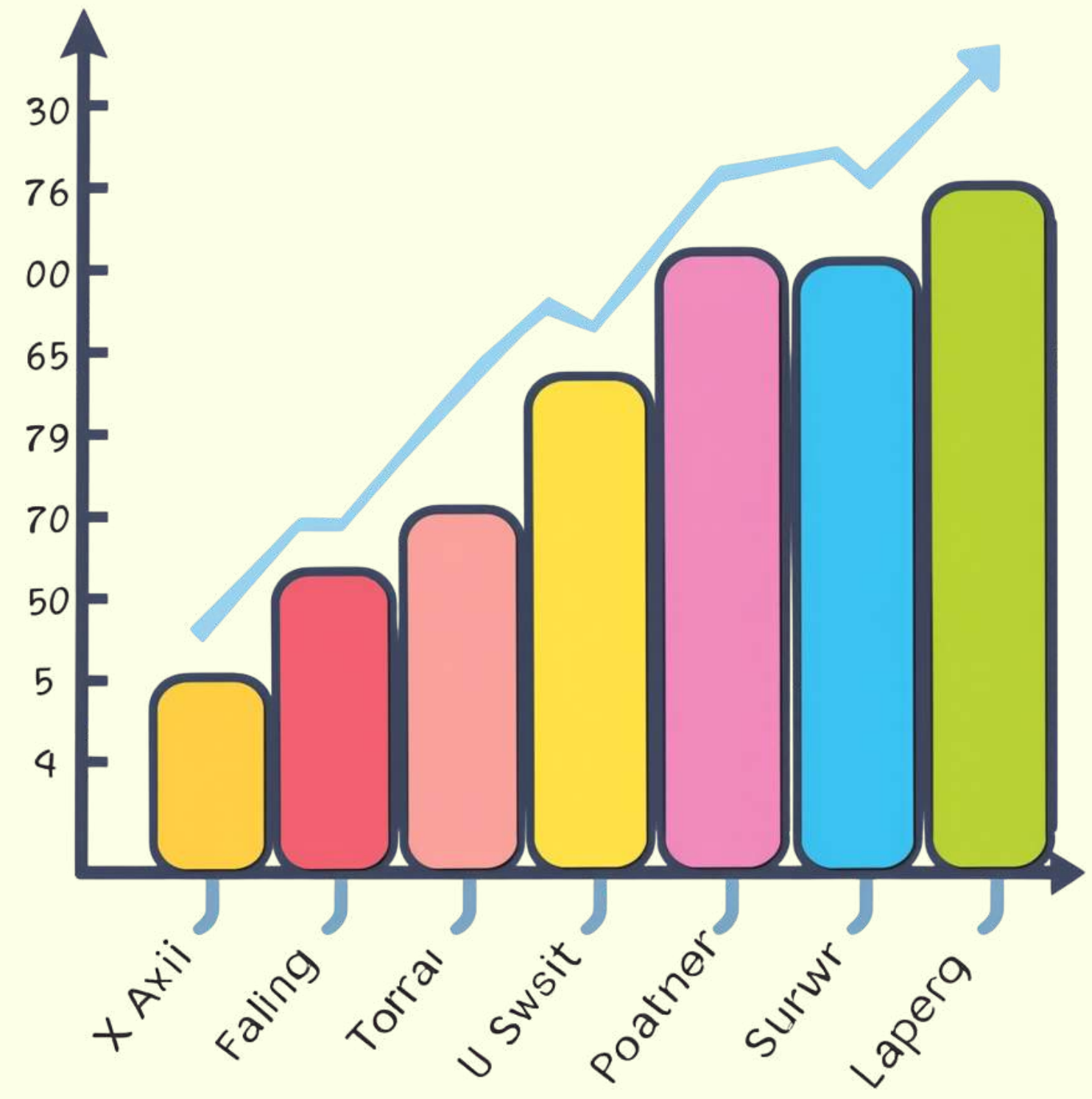
- Mumbai – 2207 mm
- Chennai – 1400 mm
- Delhi – 797 mm
- Bengaluru – 970 mm



Bar Chart Anatomy

- Title: Tells what the chart is about
- X-axis: Horizontal line with names ("X is the floor")
- Y-axis: Vertical line with numbers ("Y goes upYard")
- Bars: Rectangles showing data size (taller = bigger number)
- Values: Numbers on the Y-axis

Title BAR CHART



Reading a Bar Chart

- Q1: Which sport got the most votes? (Cricket)
Q2: How many chose Badminton? (11)
Q3: Which sport got fewer votes: Kabaddi or Football? (Football)



What is Python?

Python is a programming language that is easy to read, just like English! Big companies in India use it: ISRO, IITs, Infosys, TCS, and Flipkart. Think of Python as giving instructions to a smart robot friend!





What is Matplotlib?

Matplotlib is Python's special tool for drawing charts and graphs! Think of it as a magical geometry box that helps you make colorful pictures using code instead of pencils and rulers.

Python Lists

- Use square brackets [] to make a list
- Separate items with commas
- Text items go in quotes: "Cricket"
- Numbers have no quotes: 14
- Order matters: first name pairs with first number



Example Lists in Python

```
sports = ["Cricket", "Kabaddi",  
"Badminton", "Football", "Kho-Kho"]  
votes = [14, 9, 11, 7, 5]
```

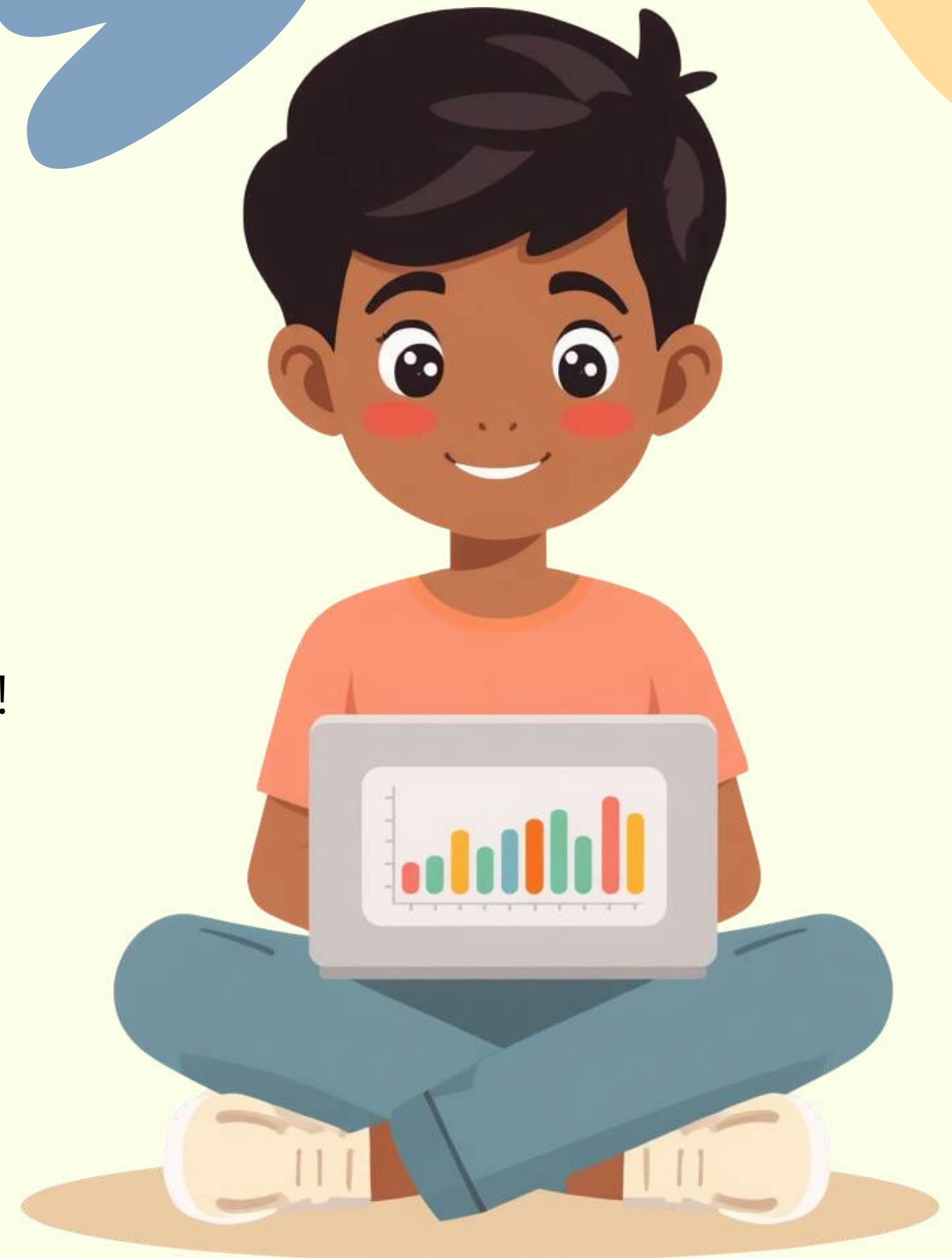


Drawing Bars with `plt.bar()`

The `plt.bar()` function takes two lists:

- First list: names (for the X-axis)
- Second list: numbers (for the Y-axis)

It draws one bar for each pair of name and number!



The 7-Line Code (Part 1)

Line 1: `import matplotlib.pyplot as plt`

Line 2: (blank line)

Line 3: `sports = ["Cricket", "Kabaddi", "Badminton", "Football", "Kho-Kho"]`

Line 4: `votes = [14, 9, 11, 7, 5]`



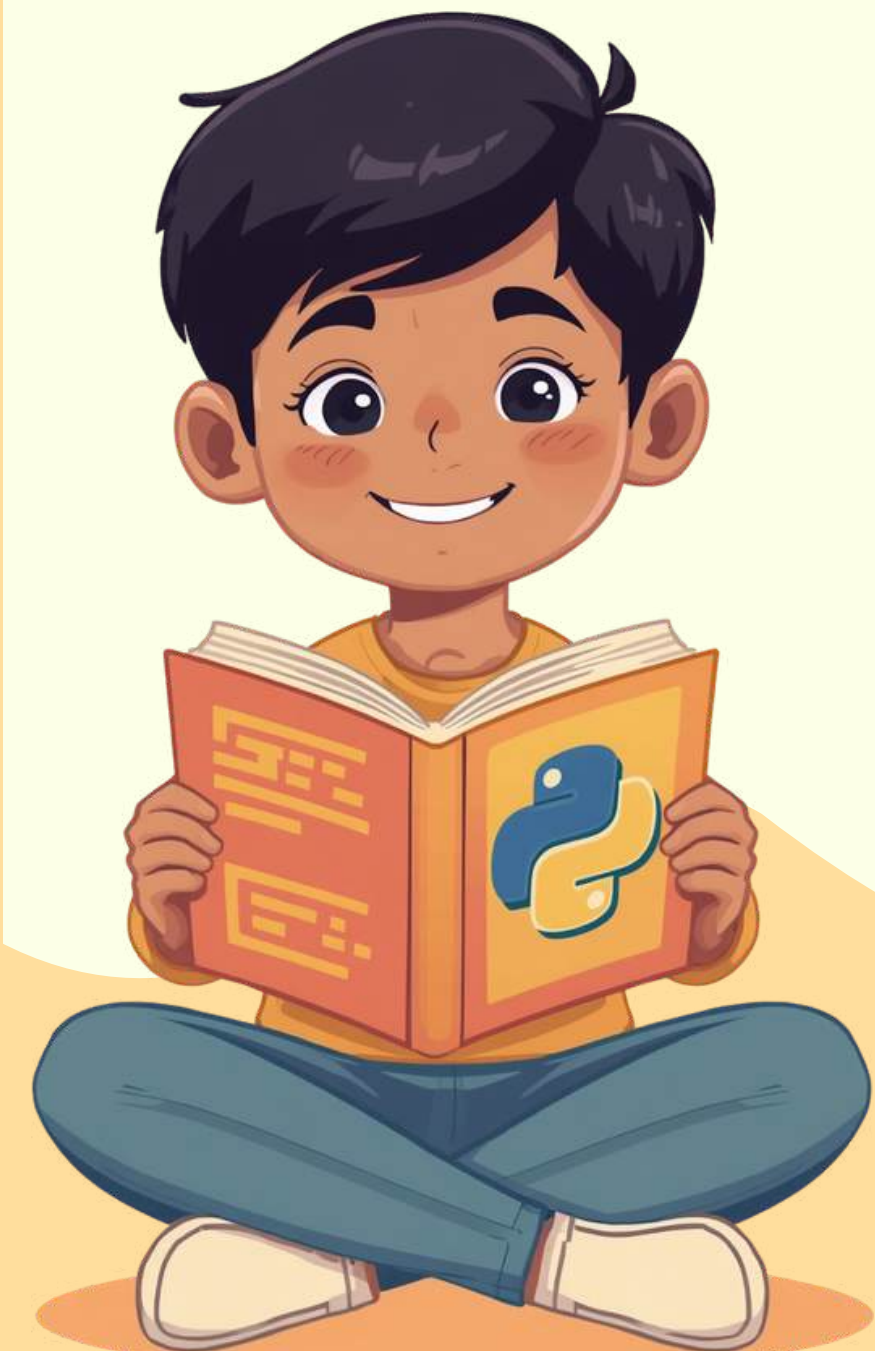
The 7-Line Code (Part 2)

Line 6: `plt.bar(sports, votes)`
Line 7: `plt.title("Favourite Sports in Our Class")`
Line 8: `plt.show()`



Code Annotations

- Line 1: Open Matplotlib toolbox and call it plt
- Line 3: List of names for X-axis
- Line 4: List of numbers for Y-axis
- Line 6: Draw bars using these lists
- Line 7: Add title to chart
- Line 8: Show the chart on screen



Common Coding Errors

Watch out for these common mistakes:

- Missing comma between list items
- Forgetting quotation marks around text
- Spelling mistakes in `plt.bar` or `plt.show`





Celebrate Success!

Great job! You made your first chart!
When your code runs correctly, it's time
to celebrate. You've learned how to turn
data into a colorful bar chart!



Challenge 1 – Change a Value

Try changing the 5 for Kho-Kho to 15 in your code!
Predict: What happens to the Kho-Kho bar?
Result: It becomes the tallest bar!



Challenge 2 – Change a Name

Try this: Change 'Football' to 'Hockey' in your code!

Predict: What will change in the chart?

Result: The label changes to Hockey, but the bar height stays the same at 7 votes!





What Did We Learn?

Challenge 1: Changing Kho-Kho from 5 to 15 made it the tallest bar!
Challenge 2: Changing 'Football' to 'Hockey' only changed the label, not the bar height!

Recap Key Concepts

- Data: What we collect
- Bar Chart: Picture comparing numbers
- X-axis: Bottom line with names
- Y-axis: Side line with numbers
- Python: Programming language
- Matplotlib: Chart tool
- `plt.bar()`: Command to draw bars





Real World Uses of Data Visualization

Data visualization helps people everywhere!

- ISRO tracks rockets and satellites
- Doctors track patient health
- Sports analysts study cricket scores

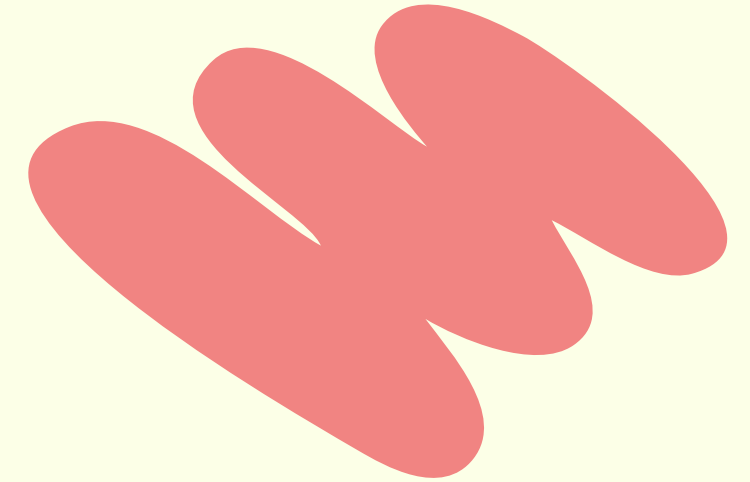
Next Session Preview

In our next session, you will:

- Collect your own data from friends and family
- Add bright colors to your bar charts
- Make your charts even cooler with customizations!



Homework Assignment



Think of something fun to count! For example:

- Favorite foods in your family
- Favorite school subjects
- Types of pets your friends have

Write down 5 names and 5 numbers to create your own data list. Bring it to class next time!





Bonus: Indian Food Survey

Let's practice with yummy Indian foods!

- Dosa – 10 votes
- Idli – 8 votes
- Roti – 12 votes (Winner!)
- Poha – 6 votes
- Upma – 7 votes



Bonus: City Rainfall Data

Look at this colorful bar chart showing how much rain falls in different Indian cities! Mumbai gets the most rain at 2207 mm, while Delhi gets the least at 797 mm. Can you spot which bar is tallest?

Bonus: School Subjects Popularity

Which subject do students love the most? Here's sample data you can use to create your own bar chart!

- Maths – 15 votes
- Science – 12 votes
- English – 9 votes
- Hindi – 11 votes
- Social Studies – 8 votes





Session Wrap-Up

You learned what data is, how to read and create bar charts, and how to write simple Python code to draw them using Matplotlib. Keep practicing and see you next time!

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Thank You!

Great job learning about data
visualization with Python!

Let's talk and explore your questions!